

Email Security Alert Gateway & Chrome Extension Installation Guide

Step-by-Step Operator Procedures for Gmail 16-Digit App Password & Chromium In-Tab Defense

Target Features:	Real-World SMTP Alerts & Browser Extension	Document Ref:	VS-DOC-2026-05
Security Protocol:	Google App Password TLS Port 587 & MV3	Supported Browsers:	Chrome, Edge, Brave, Opera

PART 1: Gmail 16-Digit App Password & Emergency Email Alert Setup

VoiceShield AI connects directly to Google's mail relay (smtp.gmail.com:587) using an encrypted TLS socket to transmit real-time emergency incident advisories to users when a synthetic cloning or cyber-extortion call is terminated. For security, Google requires an **App Password** instead of your normal account login password.

Password Type	Format & Length	Purpose & Security Rationale
Primary Account Password	User-defined (e.g. MySecret#2026)	Grants complete master access to Google account. Never share with any external application.
Google 16-Digit App Password	16 lowercase letters (e.g. abcd efgh ijkl mnop)	Restricted specifically to automated SMTP mail transmission. Can be revoked anytime with one click.

Step-by-Step Generation Procedure:

Step 1.1: Enable 2-Step Verification on Google

Google strictly requires 2-Step Verification before generating App Passwords:

- Open your browser and navigate to: <https://myaccount.google.com/security>
- Locate the section **'How you sign in to Google'**.
- Verify that **2-Step Verification** is **ON**. If OFF, click it and register your mobile number.

Step 1.2: Generate the 16-Digit App Password

- Open the direct Google App Passwords utility: <https://myaccount.google.com/apppasswords>
- Confirm your Google identity by entering your password if prompted.
- In the **'App name'** input box, type: VoiceShield AI
- Click the blue **'Create'** button.
- A popup window will display your 16-character code in a yellow box (e.g. abcd efgh ijkl mnop).
- Copy this code to your clipboard.

Step 1.3: Apply Credentials in VoiceShield AI Dashboard

1. Open the VoiceShield Web Dashboard (<http://localhost:8000> or your Cloudflare Tunnel URL).
2. Switch to **Tab 4: ■ Phone Call Defense**.
3. Scroll down to **■ Real-World Alert Gateways & Live Testing** card.
4. Enter the required parameters:
 - **Sender Gmail Address:** your_email@gmail.com
 - **Gmail 16-Digit App Password:** abcd efgh ijkl mnop
 - **Alert Recipient Email:** your_email@gmail.com (or target inbox)
5. Click **■ Save Credentials**.
6. Click  **Test Real Email to My Inbox**. VoiceShield will execute an encrypted TLS handshake to Google's server and deliver the alert.

Verification Check: Open your email inbox. You will receive an email titled:

■ [CRITICAL ALERT] Incoming Threat Call from +91-140-776655 Terminated - VoiceShield AI'
Containing formatted acoustic metrics, risk score, intercepted transcript, and cybercrime advisory.

PART 2: Installing the VoiceShield AI Chrome Browser Extension

The VoiceShield AI Browser Extension monitors incoming web audio streams—including **WhatsApp Web voice notes and calls, Google Meet, Zoom Web, and Microsoft Teams**. It intercepts voice packets in real-time, displays an on-screen HUD threat overlay, and automatically mutes synthetic deepfake audio.

Step-by-Step Installation Procedure:

Step 2.1: Open Chromium Extensions Management

Open Google Chrome (or Edge / Brave) and type the following into your top URL address bar: `chrome://extensions/` (or `edge://extensions/` on Microsoft Edge) and press **Enter**.

Step 2.2: Enable Developer Mode

In the top-right corner of the Extensions page, toggle the **'Developer mode'** switch to **ON**. Once activated, three buttons will appear in the top-left toolbar: *[Load unpacked]*, *[Pack extension]*, and *[Update]*.

Step 2.3: Load the Unpacked Extension Directory

1. Click the **'Load unpacked'** button in the top-left corner.
2. A file selection dialog will open. Navigate to the extension folder inside your repository:
`C:\SIH_2026\Voice Clone Detector\extension`
3. Click on the **extension** folder to highlight it, and click **'Select Folder'**.
4. Chrome will immediately install and activate **VoiceShield AI - Real-Time Voice Clone & Scam Blocker (v1.0.0)**.

Step 2.4: Pin the Shield Icon to the Toolbar

1. Click the **Puzzle Piece (■ Extensions)** icon located at the top-right of your Chrome window.
2. Find **VoiceShield AI** in the dropdown list.
3. Click the **Pin (■)** icon next to it so the cyan **VoiceShield Shield (■■)** icon is permanently visible.

Step 2.5: Test and Verify Extension Operation

1. Ensure your VoiceShield Python engine is running: `python run_app.py`
2. Click the **VoiceShield Shield (■■)** icon in your Chrome toolbar to open the dark-mode popup.
3. Confirm the status bar at the bottom displays: **■ Engine: Online (Conformer + Voicemod)**.
4. Under **'Forensic Engine Verification'**, select 3s Family Voice Note Clone and click **'Test Scan'**.
5. The popup will immediately turn red: **■ VOICE CLONING ATTACK DETECTED (Risk: 96.8%)**, auto-muting synthetic audio and broadcasting the HUD alert banner across active browser tabs!

Security & Privacy Guarantee:

*VoiceShield AI's Chrome extension complies with **Manifest V3**. Audio capture is performed entirely within memory buffers, never stored to browser cookies or external tracking servers, adhering strictly to the India DPDP Act 2023 zero-retention principles.*